

Vipin Neekamparambath

Agentic AI Engineer | GenAI Solution Architecture | Engineering Digitization

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Professional Summary

With 10+ years across engineering automation and digital transformation, I bring engineering-domain depth to multi-agent orchestration, retrieval-augmented generation, MCP-enabled integration, and AI copilots. My work sits where practical delivery, technical architecture, and adoption meet.

Evidence at a Glance

10+	years across engineering and transformation	5	industry roles across the engineering lifecycle	9	featured AI and automation systems	3	core AI domains: requirements, quality, simulation
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Professional Experience

Accenture DACH - Senior Analyst

Munich, Germany | Aug 2023 - Present

- Architect an Engineering Orchestrator coordinating agents across requirements, PLM, design, simulation, and downstream lifecycle workflows.
- Serve as development lead for end-to-end FMEA agents, shaping workflow decomposition, agent responsibilities, prompts, tools, and validation.
- Lead requirements engineering and requirements management agents using the Deep Agents framework.
- Develop retrieval and knowledge-management capabilities connecting critical engineering data with reusable process guidance.

Altair Engineering - Technical Specialist

Bengaluru, India | Jan 2022 - Jul 2023

- Designed, tested, and deployed tools for engineering pre-processing, post-processing, and report generation.
- Delivered end-to-end CAE automation that reduced repetitive engineering effort and improved workflow consistency.

ZF Friedrichshafen - Specialist

Hyderabad, India | Jun 2021 - Jan 2022

- Managed a centralized Azure Repos library for engineering macro collections.
- Built a Microsoft Power Apps script-library application for cross-functional teams.

Valeo - Automation Engineer

Chennai, India | Jan 2018 - Jun 2021

- Established work instructions, GUI and script templates, quality checklists, and team standards.
- Built dashboards to track specifications, delivery progress, and engineering time savings.

Tata Consultancy Services - Systems Engineer

Bengaluru, India | Nov 2014 - Dec 2017

- Created HyperWorks scripts that improved FE modelling efficiency and reinforced CAE quality standards.
- Supported vehicle subsystem assembly in HyperMesh and engineering revisions in UG NX.

Selected Agentic AI and Automation Work

Engineering Orchestrator | Agentic AI | LangChain | Azure

A multi-agent operating layer connecting requirements, PLM, design, simulation, and downstream lifecycle workflows. Impact: Created a cohesive AI-assisted layer across previously separate lifecycle activities.

FMEA Analysis Agents | LangGraph | LLM orchestration | Human review

End-to-end agents supporting workflow decomposition, context retrieval, analysis guidance, validation, and structured FMEA outputs. Impact: Made an expert workflow more consistent and easier to inspect.

Requirements Engineering Agents | Deep Agents | Requirements engineering | Tool integration

Agents assisting requirements elicitation, analysis, interpretation, management, and traceability-oriented work. Impact: Improved access to process guidance during requirements work.

Non-Conformity Agents | GenAI | RAG | Quality workflows

Quality-workflow agents helping engineers navigate context, evidence, and process-driven next actions. Impact: Made quality context and process guidance easier to navigate.

FEM Copilot | GenAI | FEM | Simulation automation

A context-aware assistant for model creation, boundary conditions, material selection, and finite-element analysis steps. Impact: Made domain guidance available closer to the modelling workflow.

Chassis Automation Tool | HyperMesh | Tcl/Tk | CAE automation

Interactive and batch automation for CAD organization, mesh creation, seam weld creation, and engineering quality checks. Impact: Reduced repetitive preprocessing effort.

Alloy Wheel DVP Automation | HyperMesh | HyperView | Tcl/Tk

An end-to-end wheel-simulation pipeline spanning pre-processing, analysis preparation, post-processing, and automated report generation. Impact: Improved wheel-modelling efficiency through end-to-end automation.

Abaqus ODB Compression Tool | Abaqus | Python | Simulation data

A standalone Python utility that reduced large Abaqus ODB result files by approximately 90% through selective variable and frame extraction. Impact: Achieved an approximately 90% compression ratio on large ODB files.

Google Slides Report Automation | Google Apps Script | Google Slides | Workflow automation

A form-driven Google Apps Script workflow that assembled engineering reports from uploaded images using a predefined Slides template. Impact: Reduced manual report assembly.

Capabilities

Agent architecture: Multi-agent orchestration | LangGraph | Deep Agents | MCP and tool integration

Knowledge and GenAI: RAG and knowledge retrieval | Prompt and workflow design | AI copilots

Engineering delivery: Requirements engineering | FMEA and quality | CAE and FEM automation | Python and Tcl/Tk | Engineering report automation

Delivery and leadership: Agile delivery | Project planning | Stakeholder management | Cross-functional collaboration | Technical leadership | Training delivery

Education and Languages

Master of Technology, Industrial Engineering & Management | National Institute of Technology Calicut | 2012 - 2014

Publication: Optimization of assembly job-shop scheduling using priority dispatching rules | CGPA: 7.4

Languages: English | Hindi | German - A1